Introduction to Typst

A modern, expressive, and blazing-fast typesetting system

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Abstract — This document serves as a comprehensive, idiomatic showcase of modern document authoring in **Typst**. We demonstrate essential capabilities including page layout control, typographic conventions, complex mathematical expressions, formatted tables, syntax-highlighted code blocks, programmatic layouts, cross-referencing, and multilingual typesetting with native Nepali (Devanagari script) support, styled in a refined dark mode aesthetic.

1. Introduction

Typst is a markup-based typesetting system designed to deliver the formatting power and precision of LaTeX alongside the intuitiveness and velocity of Markdown. Built from the ground up in Rust, Typst redefines how technical manuscripts, research papers, textbooks, and reports are created.

1.1 Architectural Advantages

Traditional typesetting tools often suffer from prolonged compile cycles, cryptic error messages, and intricate macro dependencies. Typst addresses these pain points with a clean architecture:

- **Instant Compilation**: Typst compiles documents in tens of milliseconds. With incremental rendering, edits are visible in real-time as you type.
- **Unified Paradigm**: Typst unifies text markup, mathematical typography, and code scripting into one coherent language.
- **Modern Layout Engine**: Dynamic flow layouts, multi-column regions, and pixel-perfect vector geometry are built directly into the core engine.
- **Hermetic Reproducibility**: Self-contained document execution ensures that documents produce identical results across platforms without external TeX distributions.

1.2 Syntax and Typographic Foundations

Typst features a concise, distraction-free syntax for daily writing:

- **Bold emphasis** is declared with `*bold*`
- *Italicized emphasis* is declared with `_italic_`
- `Monospaced text` is written with backticks
- Bulleted lists use simple hyphens (`-`), while ordered items use numbering (`+`)
- Footnotes can be inserted naturally anywhere in the body text[1]

[1] Typst automatically calculates footnote positioning, separators, and page numbering.

2. Mathematical Typesetting

Mathematical formulas in Typst are enclosed within dollar signs (`$`). Inline equations integrate smoothly into running text: $E = mc^2$, Euler's identity $e^{i\pi} + 1 = 0$, or the summation formula $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$.

Block equations are created by placing whitespace or linebreaks inside `$ ... $`, and automatic equation numbering is enabled effortlessly:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \tag{1}$$

We can reference this Gaussian probability distribution cleanly using Equation 1. Typst also offers concise syntax for matrix notation, linear transformations, and calculus:

$$\boldsymbol{M} = \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{2}$$

Advanced mathematical formulations, such as the continuous Fourier Transform in Equation 3, retain remarkable legibility in source markup:

$$\mathcal{F}\{f(t)\} = \hat{f}(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} \, \mathrm{d}t \tag{3}$$

Common limits and definite integrals are structured with clear semantic operators:

$$\lim_{x \to 0} \frac{\sin(x)}{x} = 1, \quad \int_0^{\infty} e^{-x^2} \, \mathrm{d}x = \frac{\sqrt{\pi}}{2}, \quad \nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} \tag{4}$$

3. Tables and Comparative Figures

Tables in Typst are declared with the `#table` function and can be wrapped in `#figure` for automated numbering, captions, and inclusion in outlines:

Evaluation Metric	Typst Approach	Traditional LaTeX
Compilation Speed	Under 100 ms (Incremental)	Multi-second multi-pass
Syntax Simplicity	Clean, Markdown-inspired	Heavy macro declarations
Package Management	Built-in Universe registry (`@preview`)	CTAN / TeX Live archive
Error Diagnostics	Precise traces with file pointers	Cryptic low-level TeX errors
Programming Model	Pure functions and closures	Macro expansions & `\def`

Table 1: Systematic comparison between Typst and traditional LaTeX typesetting workflows.

Document Size	Pages	Typst Build Time	LaTeX Build Time
Short Technical Paper	4 pages	24 ms	1,420 ms
Master Thesis	85 pages	115 ms	4,850 ms
Comprehensive Book	320 pages	340 ms	12,900 ms

Table 2: Benchmark measurements highlighting incremental build performance.

As demonstrated in Table 1 and Table 2, Typst dramatically shortens iteration cycles while maintaining uncompromising typesetting quality.

4. Syntax-Highlighted Code Listings

Typst includes built-in support for syntax highlighting across hundreds of programming languages, powered by tree-sitter. Listings blend seamlessly into the dark mode palette:

```python
def compute_fibonacci(limit: int) -> list[int]:
    """Generate Fibonacci sequence up to limit items with memoization."""
    fib: list[int] = [0, 1]
    while len(fib) < limit:
        fib.append(fib[-1] + fib[-2])
    return fib

# Compute first 10 Fibonacci numbers and display output
if __name__ == "__main__":
    sequence = compute_fibonacci(10)
    print(f"Computed sequence: {sequence}")
```

5. Document Automation and Scripting

Unlike legacy macro processors, Typst provides a full functional scripting language. You can store values in variables, define functions with `#let`, map over arrays, and generate layout elements programmatically:

Pure Rust Core (Engine)	**Tree-Sitter Syntax** (Tooling)	**Instant Watch Mode** (Workflow)
Unicode First (Typography)	**Predictable Rules** (Language)	**Zero-Config PDF** (Compiler)

6. Reusable Visual Components

Typst's composable function architecture makes it straightforward to encapsulate reusable design components, such as notices, admonitions, and callout boxes:

Compilation Tip

Execute `typst watch example.typ` in your terminal to enable automatic live re-compilation. Every time you save the file, the PDF is regenerated in milliseconds and refreshed in your viewer.

Architecture Note: Incremental Memoization

Typst's compiler uses pure function evaluations and dependency graphs. Only sections of the document affected by your latest modification are recomputed, ensuring instantaneous feedback even on large multi-chapter volumes.

7. नेपाली भाषा समर्थन र देवनागरी लिपि (Multilingual Support)

टाइपस्ट (Typst) ले युनिकोड (Unicode) र देवनागरी लिपिलाई पूर्ण रूपमा समर्थन गर्दछ। बहु-भाषिक फन्टहरूको स्वचालित व्यवस्थापन (*Font Fallback*) का कारण नेपाली तथा अन्य देवनागरी भाषाहरू कुनै अतिरिक्त जटिलता बिना स्पष्ट र सुन्दर रूपमा प्रस्तुत हुन्छन्।

नेपाली सारांश:

टाइपस्ट एक अत्याधुनिक, तीव्र र शक्तिशाली कागजात ढाँचा (typesetting) प्रणाली हो। यसले अनुसन्धान पत्र, प्रतिवेदन, पुस्तक तथा प्राविधिक लेखहरू तयार गर्न अत्यन्त सहज र प्रभावकारी वातावरण प्रदान गर्दछ। नेपाली लिपिमा जटिल संयुक्त अक्षरहरू (जस्तै: दृष्टिकोण, उत्कृष्टता, प्रविधिको) पनि शुद्ध रूपमा प्रस्तुत हुन्छन्।

अङ्क	नेपाली उखान / अभिव्यक्ति	तात्पर्य तथा अर्थ
१	नाच्न जान्दैन आँगन टेढो	आफ्नो कमजोरी ढाकछोप गर्न परिस्थितिलाई दोष दिनु
२	दुङ्गाको काप फोरेर पनि उम्रिन्छ पिपल	दृढ सङ्कल्प र लगनशीलता भए जस्तोसुकै कठिन अवस्थामा पनि सफलता प्राप्त हुन्छ
३	विद्या धनं सर्वधनं प्रधानम्	ज्ञान र विद्या नै संसारको सबैभन्दा अमूल्य र श्रेष्ठ सम्पत्ति हो

Table 3: नेपाली भाषा तथा देवनागरी लिपिमा आधारित दृष्टान्त तालिका।

8. Ecosystem and Development Workflow

Typst is supported by a rapidly growing developer ecosystem designed for seamless integration:

- **Typst CLI**: Simple command-line interface (`typst compile`, `typst watch`, and `typst init`).
- **Editor Extensions**: First-class LSP support via **Tinymist** for VS Code, Neovim, and Helix, offering auto-completion, live symbol search, and visual preview.
- **Package Universe**: Curated, community-contributed packages accessible directly via `@preview/` without manual downloads.
- **CI/CD Automation**: GitHub Actions integrate seamlessly to compile and deploy publication-ready PDFs on every commit.

9. Conclusion

Typst bridges the long-standing gap between Markdown's simplicity and LaTeX's typographic power. With instant compilation, rich mathematical typesetting, dynamic scripting, and robust internationalization, Typst stands as the next-generation standard for academic and technical communication.

References

[1] L. Haug and M. Mächler, "Typst: A Programmable Markup-based Typesetting System," 2023.

[2] D. E. Knuth, *The TeXbook*, Addison-Wesley Professional, Reading, Massachusetts, 1984.

[3] L. Lamport, *LaTeX: A Document Preparation System*, Addison-Wesley, 2nd Edition, 1994.